

Hasya Udayan Patel

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SUMMARY

Software Engineer with strong foundations in algorithms, data structures, and distributed systems. Experienced in full-stack development using React, Node.js, and Python, with hands-on projects in cloud, databases, and machine learning.

EDUCATION

Master of Computer Science with Specialization in Big Data Systems

Arizona State University, Tempe, AZ

May 2025

3.90/4.00 GPA

Courses: Software Security, Statistical Machine Learning, Data Processing at Scale, Data Mining, Data Visualization

Bachelor of Technology in Information and Communication Technology

Dhirubhai Ambani Institute of Information and Communication Technology, Gujarat, India

June 2023

3.23/4.00 GPA

Courses: Software Engineering, Cloud Computing, Computer Networks, Database Management Systems, Operating Systems

TECHNICAL SKILLS

Programming Languages: C/C++, Rust, Python, JavaScript, SQL

Web Development: React JS, Next JS, TypeScript, D3.js, Tailwind CSS, HTML, CSS, JSON

Backend and Databases: Node.js, PostgreSQL, MongoDB, Firebase

Tools: AWS, Bash, Linux/Unix, Git

Skills and Concepts: Data Structures and Algorithms, Object-Oriented Programming, Rest APIs, Full-Stack Development, Distributed Systems, Machine Learning, Agile Methodologies

PROFESSIONAL EXPERIENCE

Summer Research Intern, DAIICT, Gandhinagar, Gujarat, India

May 2022 - July 2022

Practical Homomorphic Encryption over the Integers for Secure Computation in the Cloud and its Cryptanalysis

- Implemented practical methods to validate the theoretical upper bounds of additions and multiplications for HE1 and HE1N encryption schemes.
- Developed and successfully executed a cryptographic attack on HE1 and HE1N leveraging the Lattice Reduction Algorithm.
- Engineered and optimized the OLA Attack Algorithm in **Python**, leveraging advanced polynomial optimization techniques, resulting in a 20% increase in cryptanalysis efficiency and a 15% reduction in processing time.
- Conducted extensive tests with varying polynomial degrees and entropies to assess attack time, yielding a comprehensive security assessment. Identified a 10% increase in attack time under certain conditions.

ACADEMIC PROJECT

Banking Application with Finance Management Dashboard

Spring 2025

- Built a full-stack banking application using **Next.js 14**, **React**, **TypeScript**, and **Tailwind CSS**, enabling users to connect multiple bank accounts, view real-time transactions, and initiate secure money transfers.
- Integrated Appwrite for backend services, managing user authentication, session handling, and secure API interactions.
- Implemented real-time bank account integration with Plaid API, enabling users to securely access and manage multiple accounts while providing real-time updates on transaction histories and balances.

IEEE VAST 2021 Mini Challenge 1: Data Visualization Project

Fall 2024

- Collaborated with a team to develop interactive visualizations (network timeline, word clouds, stream chart, stacked bar chart) using **D3.js** to analyze multi-source data, enhancing the exploration of narrative developments and sentiment transitions over time.
- Designed and implemented a dynamic network timeline graph to visually represent news articles' chronological progression

Disaster Tweets Classification using Natural Language Processing

Fall 2023

- Developed a machine learning model to classify tweets as disaster-related or not, achieving 81.92% accuracy with a Stacking Classifier.
- Conducted comprehensive data preprocessing, including text cleaning, tokenization, and feature extraction using TF-IDF and BERT.
- Employed models like Logistic Regression, Naive Bayes, Adaboost, SVM, Random Forest, and LSTM using libraries including **scikit-learn**, **TensorFlow**, **Keras**, **NLTK**, **NumPy**, and **Pandas**.

Graphical User Interface for Electrocardiogram (ECG) Visualization

Spring 2023

- Developed a real-time accelerometer dashboard using **Firebase** for backend support to enable live data processing and visualization.
- Designed and developed an ECG monitoring dashboard using **HTML**, **CSS**, and **JavaScript**, facilitating early detection of cardiac anomalies, and employed Flask Python framework for the backend and Python libraries like NumPy and Waveform Database (WFDB) for ECG data extraction and visualization.

EXTRACURRICULAR EXPERIENCE

IEEE IAS (Industry Applications Society) DA-IICT Student Branch Chapter: Content Writer

February 2020 – February 2021

- Documented key chapter events, including 10+ workshops and seminars, each attracting approximately 200 attendees.